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Estimating predation of the grain aphid *Sitobion avenae* by polyphagous predators

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Summary

1. The numbers of dead and live aphids falling to the ground in a cereal crop were estimated. The estimates showed that more live aphids fall to the ground than dead ones.
2. The availability of live aphids to predators which forage on the ground was low because aphids quickly returned to the crop canopy. This increased the relative availability of dead aphids.
3. Estimates of predation indicated that the staphylinids *Tachyporus* spp., which climb plants, were the most effective polyphagous predators.
4. Models comparing the potential availability of live aphids to climbing predators and ground predators indicated that predators such as *Tachyporus* spp. were able to contact more live aphid prey than were ground predators.
5. A comparison between polyphagous predators and aphid-specific hoverfly (Diptera: Syrphidae) larvae suggested that densities of syrphid larvae as low as 0.1 m^{-2} resulted in similar levels of predation to that accounted for by the polyphagous species measured in these field studies.

Key-words: aphids, polyphagous, predators, *Sitobion avenae*.

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Introduction

Polyphagous predators (staphylinid and carabid beetles, and linyphiid spiders are considered in this study) have been identified as of possible importance in the reduction of cereal aphid populations (Potts & Vickerman 1974; Wratten & Powell 1991). Enzyme-linked immunosorbent assays (ELISA) and gut dissections (Crook & Sunderland 1984; Sunderland *et al.* 1987a) have confirmed that a wide range of polyphagous predators consume the grain aphid *Sitobion avenae* (F.).

Polyphagous predators may be divided into two groups; climbing predators, including staphylinids such as *Tachyporus* spp. and the carabid *Demetrias atricapillus* (L.), and ground predators which search on the soil surface, including most carabids and linyphiid spiders, the latter building webs on or close to the ground.

This difference in the method of predation may

enable climbing predators to kill more aphids because they are potential predators of the entire aphid population, whilst ground predators consume only aphids which have fallen or walked from the crop. The availability of aphids is also influenced by the ability of the predator to respond to aggregations of prey (Bryan & Wratten 1984) and by its food preferences (Dennis, Wratten & Sotherton 1990).

Although polyphagous predators consume aphids, it is not known what proportion are scavenged or consumed alive. Only aphids which are consumed alive result in a reduction in aphid population development. This paper, using data collected during 1988 and 1989 estimates the availability of dead and live aphids to ground predators and estimates aphid predation by polyphagous predators. The differences in the potential of climbing and ground polyphagous predators to consume aphids are compared using models.

Methods

Field work was done in 1988 and 1989 at Rothamsted

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Experimental Station, UK within a 8.5-ha field of winter wheat cv. Mercia. Both trials were set up at least 50 m from field boundaries.

Aphid sampling

Aphid populations were monitored from mid May to early July in 1988 (Winder 1990) and from early May to early July in 1989 by inspecting 150 individual wheat shoots and recording the number of first to third instar nymphs, apterous and alate fourth instar nymphs and adults. The decimal crop growth stage (GS) of the crop was recorded (Zadoks, Chang & Konzak 1974). The 1988 data were used to describe the distribution of aphids between shoots by plotting the number of shoots with 0, 1, 2,, n aphids present. The frequency distribution of *S. avenae* has been described by the negative binomial distribution (Rabbinge & Mantel 1981) which has two parameters; μ , the mean number of aphids per shoot, and k , the dispersion parameter (Elliott 1977). A negative binomial distribution was fitted using the computer package MLP (Ross & Preece 1985) to the frequency distributions.

Residence time of aphids which fall to the ground

The time which aphids take to return to a wheat shoot after they have fallen to the ground was determined in the laboratory. A 1 m $\times$ 1 m tray of wheat (cv. Mercia) was sown to provide a density of 500 shoots m^{-2} , with a row spacing of 10 cm to simulate the density and spacing of field-sown winter wheat. Experiments were done during flowering.

A total of 159 first to third instar, fourth instar and adults *S. avenae* were released individually by dropping them from a height of approximately 5 cm. The instar and the residence time (α , min), the time taken to reach and start climbing a shoot, was recorded for each aphid. Three experiments were done within a growth room at constant temperatures of either 13°C or 18°C in the dark (aphids were observed in red light) and at 18°C in the light. The mean residence time was calculated for each instar.

Availability of *S. avenae* to ground predators

The availability of aphids to ground predators was estimated for 1988. A regression equation, calculated from data collected in the field (Winder 1990), with aphid density as the independent variable was used to estimate the number of aphids falling to the ground (β , aphids $\text{m}^{-2} \text{ day}^{-1}$), with a shoot density of 500 m^{-2} :

$$\beta = 500 \text{ antilog}_{10}(0.826 \log_{10}(\mu) - 0.572) \quad r = 0.78, \\ P < 0.001, n = 107$$

Assuming that living aphids were able to return to the crop canopy after α days (converted from

minutes), the mean number of living aphids present on the ground (χ , aphids m^{-2}) was estimated:

$$\chi = \alpha\beta$$

The number of dead aphids which fell to the ground was estimated by calculating the number of aphids dying due to old age and by calculating background mortality.

Carter, Dixon & Rabbinge (1982), estimated that the longevity (δ , d°) of alate adults to be 118 d° up to GS 73 and the longevity of apterous adults to be 236 d° up to GS 59 and 354 d° between GS 59 and 73 using a development threshold of -3.6°C . The number of adult aphids dying due to old age (ϕ , aphids $\text{m}^{-2} \text{ day}^{-1}$) was estimated by calculating the proportion of alate or apterous adults (ϵ , aphids shoot $^{-1}$) exceeding the maximum longevity. It was assumed that aphid ages were distributed evenly within each longevity class and that mortality occurred at a constant rate during aphid development. Mean temperature was calculated (t , °C) and mortality estimated on each day:

$$\phi = 500\epsilon(t + 3.6)/\delta$$

Dean (1974) demonstrated that during aphid development a proportion die due to a 'background' mortality (γ). Mean mortalities obtained from a range of temperatures indicated that 6.2% and 7% of apterous and alate nymphs (birth to adult moult), respectively, died during development due to background mortality (nymphal longevity being 200 d°). Adult mortality, from the pre-adult moult to maximum age was 10%.

The number of nymphs (using a mean mortality of 6.6%), and apterous and alate adult aphids dying due to background mortality (θ , aphids $\text{m}^{-2} \text{ day}^{-1}$) was estimated by calculating the proportion of nymphs, alate adults or apterous adults (ϵ) which would die during a day due to background mortality. It was assumed that the aphid population had an even age structure and that mortality occurred at a constant rate during aphid development:

$$\theta = 500\epsilon\gamma(t + 3.6)/\delta$$

Predator densities

Densities of polyphagous predator species (j) were estimated (D_j , number m^{-2}) during 1988 and 1989 by surface searching and suction sampling (Thornhill 1978). Coleopteran larvae were sampled by a suction sampler and other predators by surface searching. Densities were estimated on four occasions in 1988 and on five occasions in 1989.

Ten, random 0.1 m^2 quadrats were searched visually (for up to 15 min) and any natural enemies found were caught using a pooter. Weeds, stones, wheat plants and roots were then removed using a trowel and placed in a plastic tray. Any natural

enemies disturbed and remaining on the soil were collected. The contents of the trays were searched and natural enemies collected. Predators caught in each quadrat were frozen for later identification. Mean numbers caught per quadrat were calculated and converted to number m^{-2} .

Suction sampling was done only when the crop was dry to ensure that the suction sampler operated efficiently. On each sample date, 10 suction samples, each of five subsamples of 10 s duration were taken. Each sample was placed in a polythene bag and frozen until the contents were sorted and the predators identified. The mean number caught per sample was then calculated and converted to number m^{-2} .

Single sampling methods such as suction sampling or surface searching underestimate predator densities (Sunderland *et al.* 1987b). Their study sampled $0.1 m^2$ areas intensively by suction sampling (with a sample of 10 s duration), followed by surface searching (using a similar method to that described above) and pitfall trapping. These data were used to calculate correction factors to account for the underestimation of predator densities using a single sampling method. Factors were calculated from the proportion of predators (accounting for predators which were removed prior to surface searching by suction sampling) caught by a single method. Efficiencies were calculated for each of the 14 sites described in the study and a mean correction factor was determined for each predator group for the sampling method used. The correction factors (F_j) were 2.0, 2.1, 1.3 and 1.4 for carabids (surface searching), staphylinids (surface searching), coleopteran larvae (suction sampling) and linyphiid spiders (surface searching), respectively.

ELISA

An ELISA technique was used to determine the proportion of predators which contained remains of *S. avenae* (Crook & Sunderland 1984). For a full description of the method see Sopp (1987) and Sopp *et al.* (1992). ELISA analysis was done in 1988 and 1989 in conjunction with the predator density measurements. Test animals were collected by surface searching and pitfall trapping (over 24-h periods) within the experimental areas. Animals were frozen prior to sorting and analysis. The proportion of predators positive in an ELISA test (P_j) was calculated for each species.

Sunderland *et al.* (1987a) calculated the maximum period over which prey antigens could be detected in an individual of a given species at a constant temperature of $13^\circ C$ after they had fed to satiation (M_j , days).

Maximum consumption rates

Sopp & Wratten (1986) measured maximum laboratory consumption rates (under conditions of excess

prey) of six species of carabid and four species of staphylinid beetle. These data were used in a regression analysis to determine relationships between maximum fresh weight of aphid material eaten by species j (A_j , $mg\ day^{-1}$) and two independent variables; temperature (t , $^\circ C$) and mean weight of the predator (W_j , mg). In addition, the consumption rate of seven species of linyphiid spider and of *Tachyporus* spp. larvae were measured for this study using the same methodology. It was assumed that these laboratory consumption rates were estimates of maximum consumption rates. However, Sopp *et al.* (1992) showed that feeding rates measured in the laboratory can be lower than those recorded in an insectary when predators are allowed to actively search. Hence, these estimates may not represent the absolute maximum consumption rate. A_j was converted to the number of aphids eaten (N_j , aphids day^{-1}) by dividing by the mean aphid weight, calculated from aphid densities and the weights of individual aphid instars. Individual instar weights were obtained from Sopp (1987) and were 0.082, 0.147, 0.262, 0.455, 1.053 and 0.91 mg for first, second, third, fourth, apterous adult and alate adult *S. avenae*, respectively.

Estimation of field predation rates

Predation estimates were calculated for the weeks when predator densities and ELISA analyses were done. Upper and lower estimates of consumption of aphids (U_j and L_j , aphids $m^{-2}\ day^{-1}$) were calculated for each predator species, and it was assumed that the actual consumption was between these limits. Limits were calculated as it was not possible to calculate a single estimate of aphid predation as the effects of alternative prey, prey preferences and proportion of aphids consumed live were not known.

The upper estimate was calculated by assuming that a predator which was positive in an ELISA test had consumed live aphids at its maximum consumption rate:

$$U_j = D_j F_j P_j N_j$$

The lower estimate was calculated by assuming that a predator which was positive in an ELISA test had consumed one live aphid (of average weight) during the previous M_j days:

$$L_j = D_j F_j P_j / M_j$$

This estimate exceeded the upper estimate if one aphid/ M_j exceeded the maximum laboratory consumption rate. If this occurred the lower estimate was set to equal the upper estimate.

The proportion of predation accounted for by each predator group was calculated using a mean consumption rate calculated from the upper and lower estimates.

*Estimation of potential availability of aphids to
climbing predators and ground predators*

The potential availability of aphids to predators is defined as those that the predator could consume, given unlimited appetite and assuming each aphid requires negligible time to be consumed. Details of the estimation method are given in Appendices 1 and 2. These estimates can be compared with actual consumption to investigate whether consumption was limited by availability or by other factors. The most important assumption made was that climbing predators climb shoots at random, and that ground predators find aphids at random. The models take no account of possible modifications of behaviour in response to prey patches, such as an aggregative numerical response (Solomon 1949). The potential availability of aphids (E_j , aphids day⁻¹) to an individual of species j was estimated by

$$E_j = \left[\frac{\mu}{p_o - 1} \ln(1 - P_j) \right] / M_j$$

for climbing predators, and

$$E_j = -\ln(1 - P_j) / M_j$$

for ground predators. These equations were used to estimate potential availability during 1988.

Results*Aphid sampling*

S. avenae was the most common cereal aphid in 1988 and 1989 and results refer solely to this species. In 1988 aphids reached a maximum of 3.17 aphids per

shoot and in 1989 0.96 aphids per shoot (Table 1).

The frequency distributions of aphid populations recorded during 1988 on the 24 May, 7 & 21 June and 3 July were analysed for use with the climbing model described. Differences between the fitted and field distributions were tested using chi-squared tests, and all the fitted distributions were not significantly different from the field distributions (Fig. 1). The observed values of p_o were in good agreement with those calculated from the negative binomial, being 0.66, 0.39, 0.46 and 0.84 on 24 May, 7 June, 21 June and 4 July, respectively.

Residence times of aphids on the ground

The time taken for aphids to return to the crop canopy was longest for first to third instar nymphs and shortest for adult aphids (Table 2). Aphids took longer to return to the crop canopy in the dark but the effect of temperature was variable. All aphids returned to the crop canopy. Aphids were not selective in what they climbed, and were observed climbing leaves, shoots and stones. The mean maximum time (5.7 min) taken for aphids to return was recorded for first to third instars at 18°C in the dark (this value was used to estimate aphid availability on the ground). These estimates were similar to measurements made in a field study, in which *S. avenae* were released on the ground in the field during daylight. Field residence times of 0.79 min for alate adults, 1.1 min for apterous adults, 1.91 min for fourth instar aphids and 12.2 min for first instar aphids were obtained, with 90% of the released aphids returning to the crop canopy (Sopp 1987).

Table 1. Decimal crop growth stage (Zadoks *et al.* 1974) and population development of *S. avenae* (aphids per shoot) in winter wheat (cv. Mercia) at Rothamsted Experimental Station during 1988 and 1989

Date	GS	I–III	IV apterous	IV alate	Adult apterous	Adult alate	μ
1988							
24 May	45	1.19	0.07	0.033	0.027	0.15	1.47
27 May		1.583	0.053	0.087	0.067	0.15	1.94
31 May		1.457	0.073	0.04	0.04	0.16	1.77
3 June		1.703	0.107	0.02	0.06	0.13	2.02
7 June	58	2.723	0.167	0	0.12	0.16	3.17
10 June		2.114	0.113	0.013	0.03	0.15	2.42
14 June		2.55	0.28	0.02	0.1	0.16	3.11
17 June		2.24	0.28	0.08	0.02	0.15	2.77
21 June	70	2.073	0.12	0.087	0.04	0.19	2.51
24 June		1.607	0.093	0.02	0.04	0.12	1.88
28 June		0.899	0.067	0.007	0.007	0.05	1.03
4 July	73	0.2	0	0	0	0.007	0.21
1989							
10 May	35	0.437	0.073	0	0.02	0	0.53
22 May	47	0.367	0.033	0.02	0.033	0.027	0.48
8 June	60	0.78	0.1	0.02	0.06	0	0.96
22 June	71	0.273	0.02	0	0.007	0.02	0.32
4 July	76	0.16	0	0	0.03	0	0.19

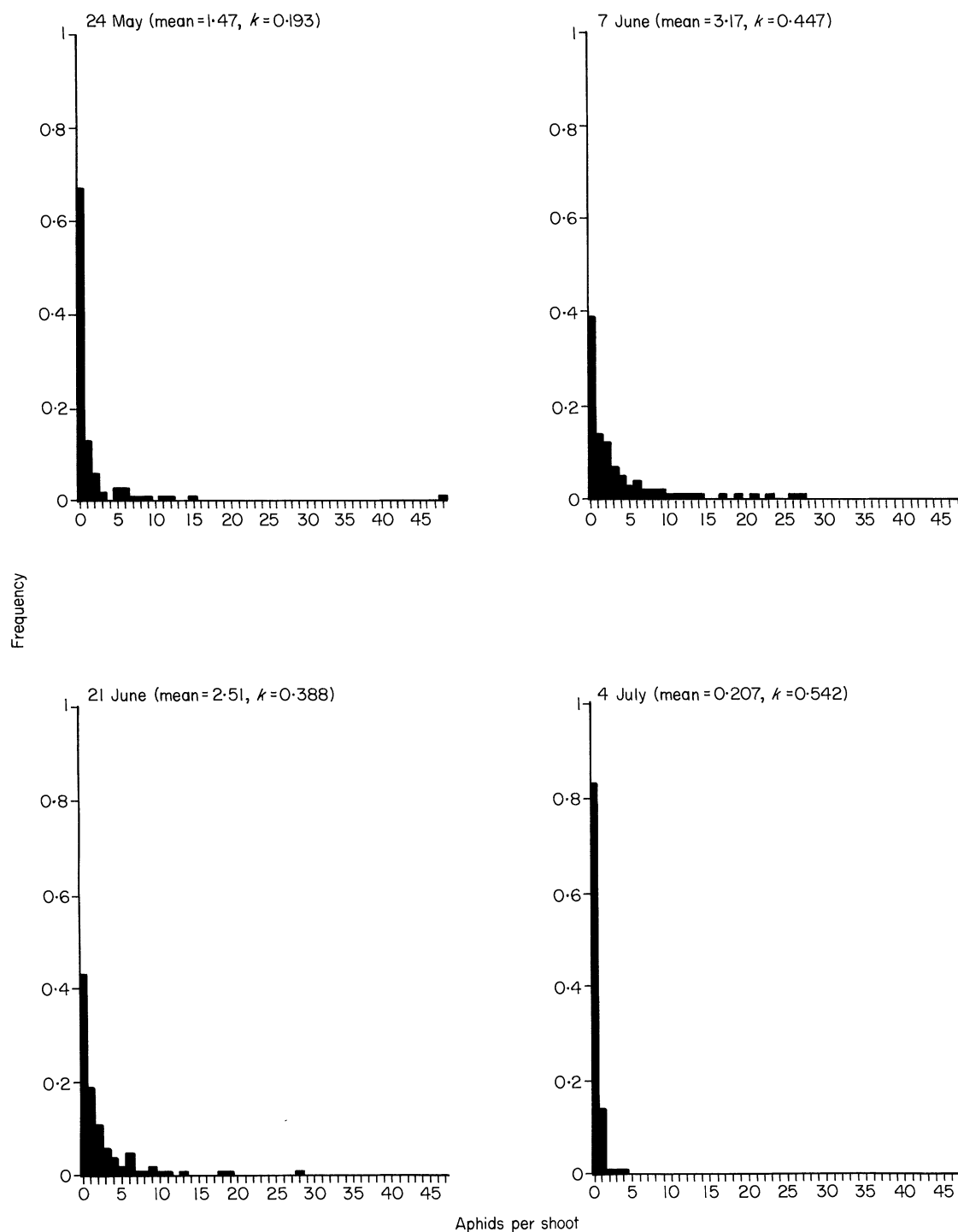


Fig. 1. Frequency distribution of *S. avenae* individuals on wheat shoots (cv. Mercia) at Rothamsted, 1988.

Table 2. Mean residence times (α , min) of *S. avenae*, measured at 18°C (light), 18°C (dark) and 13°C (dark) (number of aphids observed in parentheses)

	I-III	IV apterous	IV alate	Adult apterous	Adult alate
L18	3.2 (28)	1.1 (3)	(0)	1.7 (10)	1.5 (24)
D18	5.7 (21)	2.7 (6)	1.7 (5)	3.3 (4)	0.8 (10)
D13	4.6 (20)	4.6 (5)	3.0 (3)	2.8 (10)	1.9 (10)

Availability of aphids to ground predators

The total number of aphids falling to the ground $m^{-2} day^{-1}$, calculated from the regression equation varied from 37 on 4 July 1988 to 348 on 7 June 1988 (Table 3). The calculated mean number of aphids present on the ground m^{-2} , at any instant, varied from 0.1 on 4 July to 1.4 on 7 June.

The number of adult aphids dying $m^{-2} day^{-1}$ due to old age varied from 0.5 on 4 July to 17.4 on 21

Table 3. Availability of live and dead aphids to ground predators: mean temperature (t , °C); aphids falling to ground (β , aphids $\text{m}^{-2} \text{ day}^{-1}$); live aphids on ground (χ , aphids m^{-2}); mortality due to old age (ϕ , aphids $\text{m}^{-2} \text{ day}^{-1}$); background mortality (θ , aphids $\text{m}^{-2} \text{ day}^{-1}$)

Date	t	β	χ	ϕ	θ
1988					
24 May	12.9	184	0.7	11.5	4.7
27 May	10.6	232	0.9	11.0	5.1
31 May	12.5	215	0.8	12.3	5.4
3 June	10.5	240	0.9	9.6	5.2
7 June	10.3	348	1.4	12.9	7.9
10 June	11.2	278	1.1	10.0	6.5
14 June	13.3	342	1.3	13.9	9.3
17 June	12.5	311	1.2	10.7	8.0
21 June	16.7	287	1.1	17.4	9.4
24 June	15.2	226	0.9	10.7	6.4
28 June	13.2	138	0.5	3.8	3.1
4 July	13.5	37	0.1	0.5	0.6

June. The total number of aphids dying $\text{m}^{-2} \text{ day}^{-1}$ due to background mortality varied from 0.6 on 4 July to 9.4 on 21 June.

Predator densities

During 1988, eight species of predatory beetle were recorded (Table 4). Early in the season the beetle community was dominated by *Tachyporus* spp. As the season progressed it became dominated by *Trechus quadristriatus* (Schrank). Coleopteran larvae (predominantly *Tachyporus* spp.), and linyphiid spiders (mostly *Erigone* spp. and also *Oedothorax* spp.) were present throughout the season. During 1989 similar results were obtained, although densities were generally lower.

ELISA

In total, 879 predators were tested in 1988 and 591 in 1989 (Table 5). As the antibodies were specific to *S. avenae* (Sopp 1987) and other cereal aphids were rare during the study, it was assumed that a positive ELISA value was due to the consumption of one or more *S. avenae* over the detection period.

The collection of predators for ELISA resulted in species being caught which were not found in the quadrats used to estimate predator densities. During 1988 this included the carabids *Agonum dorsale* (Pont.), *Amara plebeja* (Gyllenhal), *Bembidion obtusum* (Serville), *D. atricapillus*, *Nebria brevicollis* (Fabr.) and *Pterostichus madidus* (Fabr.), and the staphylinids *Lathrobium fulvipenne* (Gravenhorst) and *Philonthus cognatus* (Mannerheim). During 1989 the carabids *Bembidion lampros* (Herbst.), *B. obtusum*, *Loricera pilicornis* (Fabr.), *N. brevicollis*, *P. madidus*, *Pterostichus melanarius* (Illiger) and *L. fulvipenne* were also found.

Maximum consumption rates

Significant relationships between the mean fresh weight of aphid material eaten and temperature and predator weight were derived for the four groups of predator.

Tachyporus spp. larvae:

$$A_j = 0.0721t - 0.302$$

$$r = 0.78, P < 0.001, n = 47$$

Carabidae:

$$\log_{10} A_j = 2.36 \log_{10} t + 0.495 \log_{10} W_j - 3.423$$

$$r = 0.95, P < 0.001, n = 519$$

Staphylinidae:

$$\log_{10} A_j = 1.702 \log_{10} t + 0.918 \log_{10} W_j - 2.692$$

$$r = 0.96, P < 0.001, n = 335$$

Linyphiidae:

$$A_j = 0.0170t + 0.0528W_j - 0.204$$

$$r = 0.76, P < 0.001, n = 104$$

Estimation of field predation rates

Predation rates were calculated for each week when both predator densities and a corresponding ELISA result were available. Data for carabid and staphylinid larvae were combined and total predation estimated. Linyphiids were dominated by *Erigone atra* (Blackwall) and predation was estimated using parameters for this species. Collection of invertebrates for ELISA analysis indicated that predators were present in the crop which were not caught in quadrats used for estimating predator density. The correction factor (F) was used to compensate for this.

During 1988 and 1989 *Tachyporus* spp. adults dominated aphid predation during May and early June (Figs 2 and 3, Table 6). This then declined and carabids, mainly *T. quadristriatus* became dominant. Linyphiids consumed aphids during all of the study period in 1988 and throughout most of 1989. Coleopteran larvae consumed aphids during 1988, but were present in smaller numbers during 1989 and so had less effect.

During 1988, predation estimates accounted for 3.0–4.4% of the aphid population for the week starting 23 May, 0.7–1.7% for the week starting 6 June, 0.7–4.2% for the week starting 20 June, and 14.4–123.7% for the week starting 4 July.

During 1989, predation estimates accounted for 4–5.5% of the aphid population for the week starting 7 May, 3.3–5.7% for the week starting 21 May, 1.0–1.7% for the week starting 4 June, 1.5–14.8% for the week starting 18 June, and 4.9–57% for the week starting 2 July.

Estimates of potential availability of aphids to climbing predators and ground predators

The climbing model was used to calculate estimates of aphid availability for those predator species which are known to forage on wheat shoots. Estimates of availability (for a single predator) were close to, or exceeded the maximum field predation estimates (Table 7), suggesting that consumption was not limited by availability.

The ground model was used to calculate estimates

of aphid availability for predator species which search primarily on the ground. Estimates of availability (for a single predator) were in general close to the minimum field predation estimates, suggesting that consumption may be limited by availability.

Discussion

Aphid populations did not reach damaging levels in either year. However, large numbers (Dewar & Carter 1984) of aerial aphids had been caught; 522

Table 4. Densities of polyphagous predators (D_j , number $m^{-2} \pm 1$ SE) in winter wheat (cv. Mercia) at Rothamsted during 1988 and 1989. Weights (e denotes estimate from species of similar size) of individual predators (mg) obtained from Sopp (1987; personal communication)

Mean temperature (°C)	W_j	Week starting: 1988				1989				
		23 May	6 June	20 June	4 July	7 May	21 May	4 June	18 June	2 July
		12.1	11.6	15.8	14.0	9.9	15.9	10.6	17.1	16.1
<i>Agonum dorsale</i>	17.9									4.0 ± 2.2
<i>Demetrias atricapillus</i>	4.4e					1.0 ± 1.0	1.0 ± 1.0			
<i>Bembidion lampros</i>	2.7	2.0 ± 1.3		3.0 ± 1.5						
<i>Loricera pilicornis</i>	17.9e	2.0 ± 1.3	1.0 ± 1.0		1.0 ± 1.0					
<i>Notiophilus biguttatus</i>	4.4		2.0 ± 1.3	2.0 ± 1.3	1.0 ± 1.0		1.0 ± 1.0			2.0 ± 1.3
<i>Pterostichus melanarius</i>	133.7e				1.0 ± 1.0					
<i>Trechus quadristriatus</i>	4.4e		9.0 ± 3.5	45.0 ± 10.8	77.0 ± 11.0				8.0 ± 3.3	22.0 ± 6.8
<i>Tachyporus chrysomelinus</i>	1.9	6.0 ± 4.0	5.0 ± 2.7	1.0 ± 1.0	1.0 ± 1.0	1.0 ± 1.0			1.0 ± 1.0	1.0 ± 1.0
<i>Tachyporus hypnorum</i>	2.1	6.0 ± 2.7	3.0 ± 1.5	2.0 ± 1.3		5.0 ± 2.2	4.0 ± 1.6	2.2 ± 1.5	4.0 ± 2.2	2.0 ± 2.0
<i>Tachyporus obtusus</i>	2.2	5.0 ± 2.7				3.0 ± 2.1	2.0 ± 1.3	2.2 ± 1.5	1.0 ± 1.0	
<i>Philonthus cognatus</i>	23.3							1.1 ± 1.1		
Coleopteran larvae		3.5 ± 2.2	8.3 ± 1.0	30.5 ± 4.5	14.5 ± 4.9	1.0 ± 1.0		2.2 ± 2.2	10.0 ± 3.9	3.0 ± 3.0
Linyphiids	1.8	11.0 ± 4.3	7.0 ± 5.0	10.0 ± 3.3	20.0 ± 3.7	4.0 ± 2.7	5.0 ± 1.7	10.0 ± 4.1	10.0 ± 2.6	3.0 ± 3.0

Table 5. Proportion of polyphagous predators positive in an ELISA test for *S. avenae* (P_j), (numbers in parentheses denote total individual predators tested) and the maximum detection period (M_j , days) for each species (Sunderland *et al.* 1987)

	M_j	Week starting: 1988				1989				
		23 May	6 June	20 June	4 July	7 May	21 May	4 June	18 June	2 July
<i>A. dorsale</i>	8.5	1.00(1)	0.67(3)		1.00(4)	1.00(1)	0.83(12)	0.00(1)	1.00(1)	0.83(6)
<i>D. atricapillus</i>	8.0			0.50(4)		1.00(2)				
<i>B. lampros</i>	6.5	0.23(78)	0.30(23)	0.07(28)	0.78(27)		0.00(2)			
<i>L. pilicornis</i>	5.0	1.00(7)	0.89(9)		0.75(12)	1.00(1)	0.64(11)	0.50(4)	1.00(3)	
<i>N. biguttatus</i>	3.0	1.00(8)	0.60(5)	0.38(8)	0.48(21)		0.50(2)		0.00(5)	1.00(2)
<i>P. melanarius</i>	9.0		0.00(3)	0.00(1)	0.14(4)		0.33(3)	1.00(3)	0.00(2)	
<i>T. quadristriatus</i>	5.0		0.22(9)	0.04(45)	0.26(61)				0.46(13)	0.29(41)
<i>T. chrysomelinus</i>	0.5	0.86(7)	0.63(8)	0.38(8)	0.48(21)	1.00(2)	1.00(1)			0.00(1)
<i>T. hypnorum</i>	0.5	0.89(9)	0.07(15)	0.25(8)	0.67(3)	0.88(16)	0.40(25)	0.50(2)		
<i>T. obtusus</i>	1.5	0.17(6)	1.00(1)	0.00(3)	1.00(2)		0.80(5)	0.50(2)	0.00(1)	
<i>P. cognatus</i>	1.0	1.00(3)	0.50(2)	0.00(2)	0.00(2)	1.00(2)	0.40(5)	0.11(9)	0.00(2)	
Col. larvae	3.0	0.33(3)	0.25(8)	0.28(32)	0.23(31)	1.00(1)	0.00(2)	0.00(12)	0.00(15)	0.00(3)
Linyphiids	7.0	0.32(19)	0.00(14)	0.42(24)	0.81(42)	1.00(17)	0.45(11)	0.52(23)	0.46(26)	

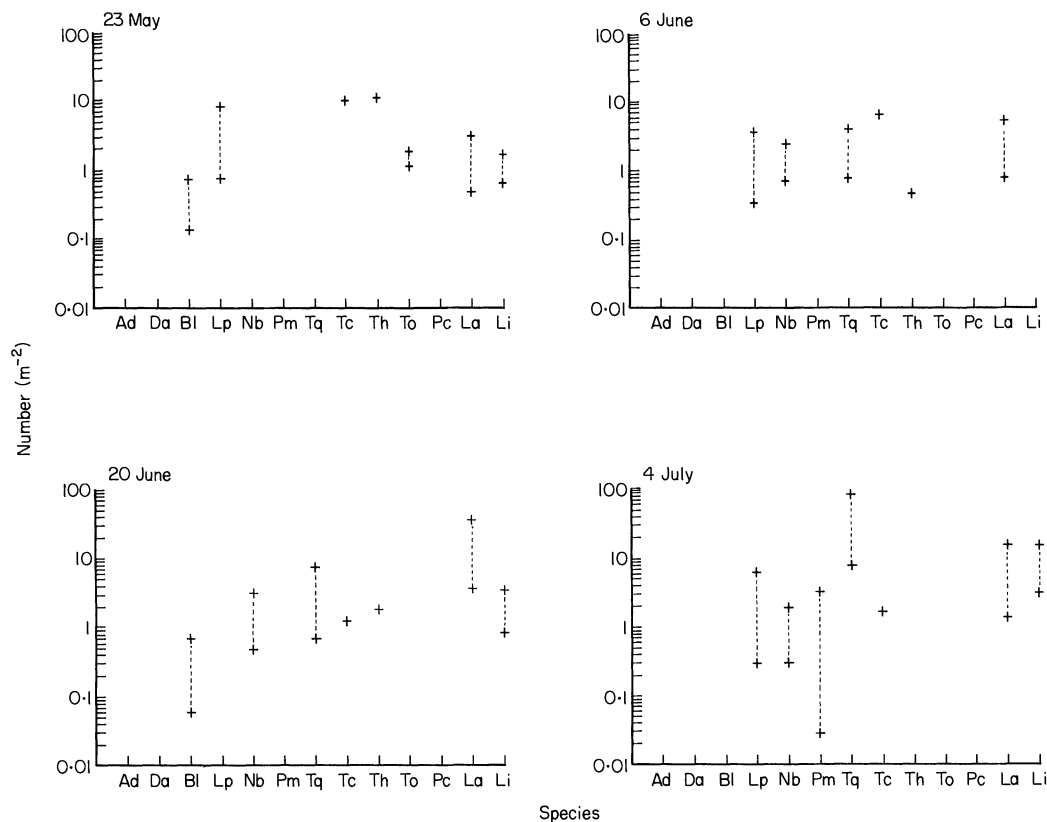


Fig. 2. Upper and lower estimates of aphid consumption (aphids $\text{m}^{-2} \text{day}^{-1}$) by polyphagous predators at Rothamsted during 1988. Ad, *Agonum dorsale*; Da, *Demetrias atricapillus*; Bl, *Bembidion lampros*; Lp, *Loricera pilicornis*; Nb, *Notiophilus biguttatus*; Pm, *Pterostichus melanarius*; Tq, *Trechus quadristriatus*; Tc, *Tachyporus chrysomelinus*; Th, *Tachyporus hypnorum*; To, *Tachyporus obtusus*; Pc, *Philonthus cognatus*; La, Coleoptera larvae, Li, *Linyphiid spiders*.

Table 6. Proportion of polyphagous predation accounted for by carabids (Ca), staphylinids (St), linyphiid spiders (Li) and coleopteran larvae (La) during 1988 and 1989

	Week starting: 1988				1989				
	23 May	6 June	20 June	4 July	7 May	21 May	4 June	18 June	2 July
Ca	0.16	0.37	0.19	0.72	0.07	0.10	0.00	0.78	1.00
St	0.73	0.42	0.10	0.02	0.74	0.80	0.77	0.00	0.00
Li	0.04	0.00	0.08	0.14	0.08	0.10	0.23	0.22	0.00
La	0.07	0.21	0.63	0.12	0.11	0.00	0.00	0.00	0.00

and 328 alate aphids were caught in the Rothamsted Insect Survey suction trap (Taylor 1977) prior to flowering during 1988 and 1989, respectively. It is likely that in the absence of mortality factors, such as natural enemies, numbers of aphids would have been greater and may have caused crop damage.

Estimates of aphid fall-off and background and old-age mortality indicated that most aphids which fell to the ground were live aphids (Table 3). However, laboratory estimates of residence times indicated that live aphids were able to return to the crop canopy rapidly (5.7 min). Once the residence time was accounted for, the availability of aphids to

ground predators was considerably reduced from that suggested by aphid fall-off. This therefore increases the relative availability of dead aphids to ground predators. Dead aphids remain on the ground until they are either scavenged or decompose. An individual dead aphid is therefore likely to be exposed for longer to ground predators than a live aphid which returns to the crop. As ground predators have the opportunity to scavenge as well as consume live aphids, a predator positive in an ELISA test does not necessarily indicate a role in aphid population control.

Differences in predator searching behaviour may affect the relative availability of dead and live aphids. Predation by predators which do not scavenge, such as linyphiid spiders (Fraser 1982; Sunderland *et al.* 1987a) or lycosid spiders (M. Thomas, personal communication) will consume only living aphids. In the case of spiders, mortality of aphids may occur due to webs which can cover 50% of a field (Sunderland, Fraser & Dixon 1986). Such webs can kill small aphids even when the spider is absent. A proportion of aphids which fall to the ground may be intercepted by these webs and cause mortality. Predators which forage within the crop canopy, such as *Tachyporus* spp. (Holmes 1983; Dennis *et al.* 1990), coleopteran larvae (Holmes 1983) and

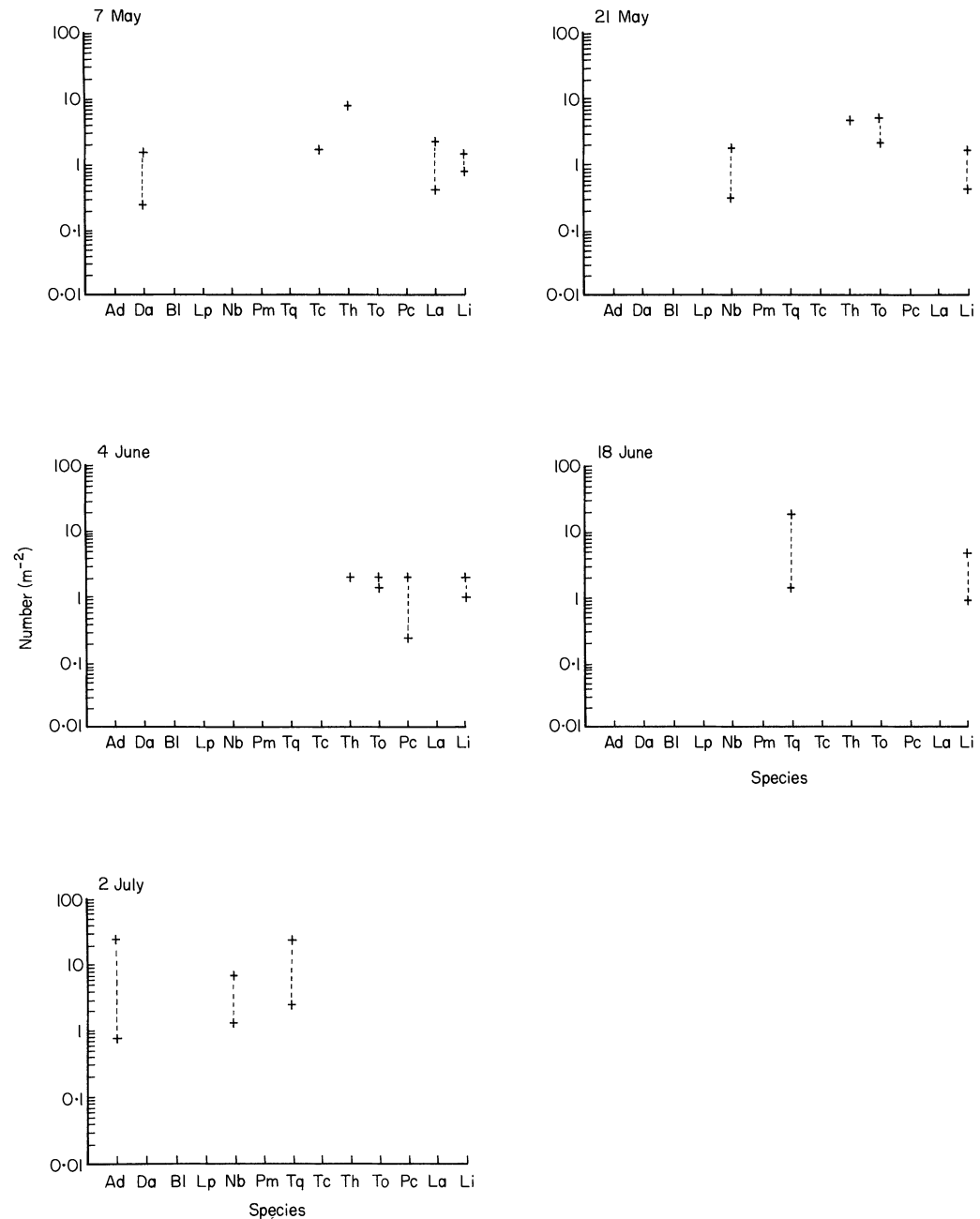


Fig. 3. Upper and lower estimates of aphid consumption (aphids $\text{m}^{-2} \text{day}^{-1}$) by polyphagous predators at Rothamsted during 1989.

D. atricapillus (Coombes 1987) may eat more live aphids as all the population is accessible when compared to a similar species restricted to the ground. Predators which search only on the ground (including most carabid beetles) and also scavenge are likely to have the smallest impact on aphid population growth.

Predation estimates (which assumed that there was no scavenging) indicated that staphylinid and carabid adults, coleopteran larvae and linyphiid spiders all consumed aphids. Staphylinid adults, particularly *Tachyporus* spp. accounted for most of the predation by polyphagous predators, particularly early in the season. Polyphagous predators are

considered to be important at this time as other natural enemies such as syrphids invade the crop later in the season.

The models of predation indicate that live aphid availability is much greater for climbing predators than ground predators. The estimates of potential availability of aphids was close to, or exceeded the maximum actually consumed for all climbing predators. This suggested that predation by these species was not limited by availability, which in turn implies that total predation is unlikely to increase greatly if the number of aphids increases. The estimates of potential availability for ground predators, however, are always close to the minimum actually

Table 7. Potential availability estimates (E_j , aphids day⁻¹) of *S. avenae* for climbing (c) and ground (g) predators observed during 1988. Also shown are upper (u_j , aphids day⁻¹) and lower (l_j , aphids day⁻¹) predation estimates calculated for individual predators

		23 May			6 June			20 June			4 July		
		l_j	E_j	u_j	l_j	E_j	u_j	l_j	E_j	u_j	l_j	E_j	u_j
<i>A. dorsale</i>	g				0.08	0.13	1.43						
<i>D. atricapillus</i>	c							0.06	0.4	1.06			
<i>B. lampros</i>	g	0.04	0.04	0.19	0.05	0.05	0.25	0.01	0.01	0.12	0.12	0.23	1.28
<i>L. pilicornis</i>	g				0.18	0.44	1.87				0.15	0.28	3.20
<i>N. biguttatus</i>	g				0.2	0.31	0.63	0.13	0.16	0.80	0.16	0.22	1.02
<i>P. melanarius</i>	g										0.02	0.02	1.59
<i>T. quadristriatus</i>	g				0.04	0.05	0.23	0.01	0.01	0.08	0.05	0.06	0.55
<i>T. chrysomelinus</i>	c	0.81	17.0	0.81	0.61	10.3	0.61	0.60	4.44	0.60	0.83	1.69	0.83
<i>T. hypnorum</i>	c	0.92	19.1	0.92	0.07	0.75	0.07	0.44	2.67	0.44	1.26	2.87	1.26
<i>T. obtusum</i>	c	0.12	0.54	0.18									
<i>P. cognatus</i>	c				0.5	3.6	4.94						
Col. larvae	c	0.11	0.58	0.7	0.08	0.5	0.56	0.09	0.51	0.94	0.08	0.11	0.86
Linyphiids	g	0.05	0.06	0.11				0.06	0.08	0.27	0.12	0.24	0.55

Table 8. Mean number of third instar *S. avenae* eaten (aphids day⁻¹) by the larvae of *Episyrphus balteatus* (Ankersmit *et al.* 1986) with a fluctuating temperature regime, 21–13°C (mean temperature 17°C) and density (number m⁻²) of *E. balteatus* of each age (days) required to be equivalent to the mean observed polyphagous predation at Rothamsted during 1988 and 1989

		Week starting: 1988				1989				
Age (days)	Aphids eaten	23 May	6 June	20 June	4 July	7 May	21 May	4 June	18 June	2 July
1	0.9	36.1	17.0	34.3	57.8	12.8	13.9	7.1	13.7	38.1
2	2.0	16.2	7.7	15.4	26.0	5.7	6.3	3.0	6.2	17.2
3	3.8	8.5	4.0	8.1	13.7	3.0	3.3	1.7	3.2	9.0
4	10.2	3.2	1.5	3.0	5.1	1.1	1.2	0.6	1.2	3.4
5	15.3	2.1	1.0	2.0	3.3	0.7	0.8	0.4	0.8	2.2
6	51.6	0.6	0.3	0.6	1.0	0.2	0.2	0.1	0.2	0.7
7	48.0	0.7	0.3	0.6	1.1	0.2	0.3	0.1	0.3	0.7
8	32.8	0.9	0.5	0.9	1.6	0.4	0.4	0.2	0.4	1.1
9	0.4	81.1	38.3	77.0	130.0	28.7	31.3	15.9	30.8	85.7

consumed. This suggests that availability may be a limiting factor and that total consumption could be increased considerably if aphid numbers increased. However, the total number of aphids available to ground predators can only be a relatively small proportion of the aphid population due to their short residence time on the ground. The estimates of potential availability to ground predators assumed that predators foraged randomly. This assumption was included because the estimate of live aphid availability to ground predators was low (Table 3), thus making the formation of a prey patch at the observed aphid densities unlikely. Bryan & Wratten (1984), however, demonstrated that *A. dorsale*, *A. plebeja*, *B. lampros*, *B. obtusum*, *P. cognatus*, *T. chrysomelinus*, *T. hypnorum*, *Tachyporus nitidulus* (Fabr.), *T. obtusus* and *Tachinus rufipes* (DeGeer) could respond to patches of *S. avenae*, in tests where

aphid densities were manipulated. The study did not identify whether predators responded to honeydew, aphids within the crop canopy, dead aphids on the ground, or live aphids on the ground. However, pitfall-trap catches of aphids were large in the 'patch' areas, suggesting soil surface activity by this group.

Other natural enemies such as syrphids, parasitoids and coccinellids consume only living aphids and therefore contribute to aphid control. The relative importance of each predator group is unclear. The syrphid *Episyrphus balteatus* (Deg.) is found in cereal crops in southern England and is considered to be important in some years (Chambers & Adams, 1986). A comparison of the laboratory aphid consumption rate of *E. balteatus* (Ankersmit *et al.* 1986) and polyphagous predation estimated in this study suggested that the predation by all polyphagous predators could be accounted for by *E. balteatus*

larval densities as low as 0.1 m^{-2} (Table 8) assuming that the laboratory estimates are close to field consumption rates. This assumption may be incorrect. For example, differences have been observed in the consumption of *S. avenae* by *Coccinella 7-punctata* (L.) between the laboratory (in small Ankersmit-type seedling pots) and large field cages (Whittaker 1987). However, the comparison still emphasizes the possibility that natural enemies such as syrphids, even at low densities may swamp the effects of polyphagous predators. The difficulty in estimating numbers of predators when densities are very low may have resulted in the role of voracious predators such as syrphids being underestimated. Syrphid larvae are active at night so this compounds the sampling problems for this group.

Winder (1990) demonstrated that aphids returned to the crop canopy at different rates when predator densities were manipulated using polythene barriers. Aphids returned to the crop canopy at a lower rate when predator densities were highest, and it was suggested that this may have resulted in smaller aphid populations developing. However, the results summarized above indicate that the study did not measure densities of natural enemies such as syrphids with sufficient precision to discount them completely from causing the observed differences.

Until the role of all natural enemy groups are measured at the same time with sufficient accuracy, the relative importance of each predator group will remain unclear.

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Appendix 1

Estimating the potential number of aphids available to climbing predators

The potential number of aphids available is defined as those that could be eaten if the predator had unlimited appetite and took no time to consume each aphid. Assume a predator climbs shoots at random, climbing X shoots in time t where $X \sim \text{Poisson}(\lambda)$. Suppose λ is constant for all predators from the same species. Suppose the number of aphids on the i th shoot has a negative binomial distribution, $Y_i \sim \text{NB}(k, \mu)$, where μ is the mean and k the dispersion parameter, independent of the others climbed. Suppose all of the aphids on the shoot are available, and also that the time taken between shoots is independent of the number of aphids on a shoot. Also assume that a predator never encounters the same aphid more than once, and that no two predators encounter the same aphid. Let the total number of aphids available to the predator in time t be Z where $Z = Y_1 + \dots + Y_x$.

Standard results now give the mean and variance of Z as

$$\begin{aligned} E(Z) &= E(X)E(Y) \\ &= \lambda\mu \end{aligned}$$

and

$$\begin{aligned} \text{var}(Z) &= E(X)\text{var}(Y) + E(Y)^2\text{var}(X) \\ &= \lambda\mu(1 + \mu + \mu/k) \end{aligned}$$

Parameters μ and k are estimated from field data; λ can be estimated by assuming that a predator will only be negative in an ELISA test if it has not encountered any aphids in the previous M_j days, in other words that it will consume at least one of the aphids available in the period. We then equate the proportion of predators which did not consume aphids over M_j days with $\text{prob}(Z=0)$. Let this proportion be $1 - P_j$, and let $p = k/(k + \mu)$. Then

$$\begin{aligned} 1 - P_j &= p(X=0) + p(X=1 \wedge Y_1=0) + \\ &\quad p(X=2 \wedge Y_1=0 \wedge Y_2=0) + \dots \\ &= e^{-\lambda} + \lambda e^{-\lambda} p^k + \frac{\lambda^2 e^{-\lambda} p^{2k}}{2} + \dots \\ &= e^{-\lambda} \left[1 + \sum_{i=1}^{\infty} \frac{(\lambda p^k)^i}{i!} \right] \end{aligned}$$

But $e_x = 1 + x + x^2/2! + x^3/3! + \dots$ and so

$$\begin{aligned} 1 - P_j &= e^{-\lambda} e^{\lambda p^k} \\ &= e^{\lambda(p^k - 1)} \end{aligned}$$

$$\text{i.e. } \lambda = \frac{\ln(1 - P_j)}{p^k - 1}$$

But note that $p^k = \text{prob}(\text{no aphids on a shoot}) = p_0$, say, and so

$$E(Z) = \frac{\mu}{(p_0 - 1)} \ln(1 - P_j)$$

The mean number of aphids available per day to a predator of species j is therefore $E(Z)/M_j$ and the variance is $\text{var}(Z)/M_j^2$.

Appendix 2

Estimating the potential number of aphids available to non-climbing predators

Assume predator from species j encounters aphids at random and that the probability of a predator encountering an aphid is equal for all predators from a given species. Also assume that there is no clustering of aphids on the ground within the area searched by the predators. The number of aphids available in time t is then Z where $Z \sim \text{Poisson}(\lambda)$ and λ is constant for all predators from a given species. The probability of a predator encountering no aphids is $p(Z=0) = e^{-\lambda}$. Equating this with the proportion $1 - P_j$ of predators found not to have consumed aphids in M_j days gives the following estimate:

$$\lambda = -\ln(1 - P_j)$$

This is equal to both the mean and variance of the Poisson distribution, and so the mean number of aphids available per day to a predator from species j is

$$E(Z) = -\ln(1 - P_j)/M_j,$$

and the variance is

$$\text{var}(Z) = -\ln(1 - P_j)/M_j^2$$